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Technical Specification

3rd Generation Partnership Project; Technical Specification Group Services and System Aspects; 5G Security Assurance Specification (SCAS); Network Data Analytics Function (NWDAF) (Release 20)





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Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

Version x.y.z

where:

x the first digit:

1 presented to TSG for information;

2 presented to TSG for approval;

3 or greater indicates TSG approved document under change control.

y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.

z the third digit is incremented when editorial only changes have been incorporated in the document.

In the present document, modal verbs have the following meanings:

shall indicates a mandatory requirement to do something

shall not indicates an interdiction (prohibition) to do something

The constructions "shall" and "shall not" are confined to the context of normative provisions, and do not appear in Technical Reports.

The constructions "must" and "must not" are not used as substitutes for "shall" and "shall not". Their use is avoided insofar as possible, and they are not used in a normative context except in a direct citation from an external, referenced, non-3GPP document, or so as to maintain continuity of style when extending or modifying the provisions of such a referenced document.

should indicates a recommendation to do something

should not indicates a recommendation not to do something

may indicates permission to do something

need not indicates permission not to do something

The construction "may not" is ambiguous and is not used in normative elements. The unambiguous constructions "might not" or "shall not" are used instead, depending upon the meaning intended.

can indicates that something is possible

cannot indicates that something is impossible

The constructions "can" and "cannot" are not substitutes for "may" and "need not".

will indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

will not indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

might indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

might not indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

is (or any other verb in the indicative mood) indicates a statement of fact

is not (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

1 Scope

The present document contains requirements and test cases that are specific to the NWDAF network product class. It refers to the Catalogue of General Security Assurance Requirements and formulates specific adaptions of the requirements and test cases, as well as specifying requirements and test cases unique to the NWDAF network product class.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 23.288: "Architecture enhancements for 5G System (5GS) to support network data analytics services".

[3] 3GPP TS 33.117: "Catalogue of general security assurance requirements".

[4] 3GPP TR 33.926: "Security Assurance Specification (SCAS) threats and critical assets in 3GPP network product classes".

[5] 3GPP TS 23.501: "System Architecture for 5G System (5GS)".

[6] 3GPP TS 29.552: "Network Data Analytics signalling flows".

[7] 3GPP TS 33.501: "Security architecture and procedures for 5G System".

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in TR 21.905 [1].

3.2 Symbols

Void

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in TR 21.905 [1].

4 NWDAF-specific security requirements and related test cases

4.1 Introduction

NWDAF specific security requirements include both requirements derived from NWDAF-specific security functional requirements in relevant specifications as well as security requirements introduced in the present document derived from the threats specific to NWDAF as described in TR 33.926 [4].

4.2 NWDAF-specific security functional requirements and related test cases

4.2.1 Technical baseline

4.2.1.1 General

The present clause provides baseline technical requirements.

4.2.1.2 Protecting data and information

4.2.1.2.1 Protecting data and information – general

There are no NWDAF-specific additions to clause 4.2.3.2.1 of TS 33.117 [3].

4.2.1.2.2 Protecting data and information – Confidential System Internal Data

There are no NWDAF-specific additions to clause 4.2.3.2.2 of TS 33.117 [3].

4.2.1.2.3 Protecting data and information in storage

There are no NWDAF-specific additions to clause 4.2.3.2.3 of TS 33.117 [3].

4.2.1.2.4 Protecting data and information in transfer

There are no NWDAF-specific additions to clause 4.2.3.2.4 of TS 33.117 [3].

4.2.1.2.5 Logging access to personal data

There are no NWDAF-specific additions to clause 4.2.3.2.5 of TS 33.117 [3].

4.2.1.2.6 Protecting data and information – Data masking on integration analysis

Requirement Name: Data masking on integration analysis about personal data

Requirement Reference: In accordance with industry best practice..

Requirement Description: NWDAF can collect data from OAM, MDAF and/or 5GC NFs (e.g. AMF) etc. for analytics purposes. Since personal data of the users could be involved , there is a potential privacy impact. As the NWDAF can expose its service operations with a request for bulked data, anonymization of data fields shall be applied to avoid exposing undesired information, aggregation levels.

Threat References: TR 33.926 [4], clause 5.3.6.7, Personal Identification Information Violation

Test Name: TC_DATA_MASKING

Purpose:

Verify that no privacy-related information of the subscribers is disclosed to any entity who is not authorized to access such information.

Pre-Condition:

Privacy information list (contains e.g. PII, location data, network identifiers, session information; should be specified based on local policy, regulation and others).

NOTE: If user consent check is implemented, user consent for data collection is granted.

The following entities are operational, integrated and simulated:

- NWDAF.
- 'data producer' (NF- or OAM as source for data collection which generates user data containing privacy info, e.g. AMF).
- 'analytics consumer' (NF- or OAM to which the NWDAF exposes analytics).

The data producer is configured to receive and accept subscription requests from the NWDAF for events according to TS 29.552 [6], clause 5.5.1.1.

Execution Steps:

1. Tester triggers behaviour so that the 'data producer' is required to handle privacy information (e.g. for AMF trigger registration request at UE).
2. The tester sends an Nnwdaf_AnalyticsInfo_Request request message from the 'analytics consumer' to NWDAF according to TS 29.552 [6], clause 5.2.3.1. The request message shall be crafted to capture information from step 1.
3. The tester retrieves the Nnwdaf_AnalyticsInfo_Request response message from the NWDAF.

Expected Results:

The analytics results do not reveal subscriber permanent identifier nor any other data listed on the Privacy information list.

Expected format of evidence:

Evidence suitable for the interface, e.g. screenshot, pcap trace, log files containing the results.

4.2.1.2.7 Fine-grained authorization for analytics services based on Analytics ID

Requirement Name: Fine-grained authorization for analytics services

Requirement Reference: In accordance with TS 33.501 [7], clauses X.8, X.9, and X.12.

Requirement Description: The NF Service Producer authenticates the NF Service Consumer and verifies the access token and ensures that the Analytics ID is included in the access token.

Threat References: TR 33.926 [4], threats related to "Privilege Escalation" and "Unauthorized Service Access".

Test Name: TC_NWDAF_ANALYTICS_ID_AUTHZ

Purpose:

To verify that the NWDAF, when acting as an NF service producer, can perform fine-grained authorization control based on the Analytics ID claim in an access token within procedures such as Nnwdaf_RoamingAnalytics_Subscribe/Request. The test ensures that a service consumer is only able to access the specific analytics type(s) it is authorized for.

NOTE: This test case applies only when the access token includes Analytics ID.

Pre-Conditions:

- The NWDAF under test is configured to support at least two distinct analytics services, Analytics ID 1 and Analytics ID 2.
- The tester has the capability to generate valid OAuth 2.0 access tokens. Two valid OAuth 2.0 access tokens are pre-configured and prepared for testing:
  - Token_A: Contains Analytics ID 1 in its claims.
  - Token_B: Contains Analytics ID 2 in its claims.
- The vendor’s product documentation describes the NWDAF's authorization policies and the APIs for the analytics services.
- The NWDAF under test has already authenticated the NF service consumer (simulated by the tester).

Execution Steps:

1. The tester sends an Nnwdaf_RoamingAnalytics_Subscribe/Request message to the NWDAF for Analytics ID 1, including Token_A in the request. The tester observes and records the NWDAF's response.
2. The tester sends an Nnwdaf_RoamingAnalytics_Subscribe/Request message to the NWDAF for Analytics ID 2, including Token_A in the request. The tester observes and records the NWDAF's response.
3. The tester repeats step 1, but uses Token_B for the request to Analytics ID 1.
4. The tester repeats step 2, but uses Token_B for the request to Analytics ID 2.

Expected Results:

1. The request in Step 1 (authorized Analytics ID) is processed successfully by the NWDAF (subject to other normal conditions being met).
2. The request in Step 2 (unauthorized Analytics ID) is rejected by the NWDAF with an appropriate error (e.g., 403 Forbidden with OAuth 2.0 error insufficient_scope).
3. The request in Step 3 (unauthorized Analytics ID for Token_B) is rejected by the NWDAF with an appropriate error.
4. The request in Step 4 (authorized Analytics ID for Token_B) is processed successfully by the NWDAF.

Expected format of evidence:

Logs, network traces (e.g., .pcap files), or screenshots of the relevant interface messages. The evidence must clearly show:

1. The Nnwdaf_RoamingAnalytics_Subscribe/Request request messages, including the target Analytics ID and the access token used (Token A or Token B).
2. The corresponding success or error response (including status code and error details) from the NWDAF.

4.2.1.2.8 ML model asset access control in cross-NWDAF model provisioning

Requirement Name: Analytics ID-level access control for ML model assets in MTLF provisioning

Requirement Reference: In accordance with TS 33.501 [7], clauses X.9 and X.10.

Requirement Description: The NWDAF containing the MTLF verifies that the Analytics ID contained within the access token's scope matches the Analytics ID associated with the requested ML model asset.

Threat References: TR 33.926 [4], threats related to "Unauthorized Access to Critical Assets" and "Model Poisoning".

Test Name: TC_NWDAF_MLMODEL_ACCESS_CONTROL

Purpose:

To verify that the NWDAF, when acting as an MTLF (NF service producer), can perform fine-grained authorization control based on the Analytics ID claim in an access token within the Nnwdaf_MLModelProvision_Subscribe procedure. The test ensures that a service consumer is only able to access the specific ML model asset(s) it is authorized for.

NOTE: This test case applies only when the access token includes Analytics ID.

Pre-Conditions:

- The vendor's product documentation describes the ML model provisioning procedure and the associated authorization mechanism for the Nnwdaf_MLModelProvision_Subscribe service operation.
- A test environment with at least one NWDAF containing an MTLF and one NWDAF containing an AnLF (or an equivalent simulation provided by the vendor) shall be available.
- The MTLF under test is configured with at least two distinct ML model assets, identifiable by different Analytics IDs (e.g., Analytics ID 1 and Analytics ID 2).
- The tester has the capability to generate valid OAuth 2.0 access tokens. Two valid OAuth 2.0 access tokens are pre-configured and prepared for testing:
  - Token_A: Contains Analytics ID 1 in its claims.
  - Token_B: Contains Analytics ID 2 in its claims.

Execution Steps:

1. The tester sends an Nnwdaf_MLModelProvision_Subscribe request to the MTLF for the ML model corresponding to Analytics ID 1, including Token A in the request. The tester observes and records the MTLF's response.
2. The tester sends an Nnwdaf_MLModelProvision_Subscribe request to the MTLF for the ML model corresponding to Analytics ID 2, including Token A in the request. The tester observes and records the MTLF's response.
3. The tester repeats step 1, but uses Token_B for the request to Analytics ID 1.
4. The tester repeats step 2, but uses Token_B for the request to Analytics ID 2.

Expected Results:

1. The request in Step 1 (authorized Analytics ID for Token A) is accepted by the MTLF, and ML model information for Analytics ID 1 is returned normally.
2. The request in Step 2 (unauthorized Analytics ID for Token A) is rejected by the MTLF with an appropriate error (e.g., 403 Forbidden with OAuth 2.0 error insufficient_scope).
3. The request in Step 3 (unauthorized Analytics ID for Token B) is rejected by the MTLF with an appropriate error.
4. The request in Step 4 (authorized Analytics ID for Token B) is accepted by the MTLF, and ML model information for Analytics ID 2 is returned normally.

Expected format of evidence:

Logs, network traces (e.g., .pcap files), or screenshots of the relevant interface messages. The evidence must clearly show:

1. The Nnwdaf_MLModelProvision_Subscribe request messages, including the target Analytics ID and the access token used (Token A or Token B).
2. The corresponding success or error response (including status code and error details) from the MTLF for each of the four execution steps.

4.2.2 Void

4.3 NWDAF-specific adaptations of hardening requirements and related test cases

4.3.1 Introduction

The present clause contains NWDAF-specific adaptations of hardening requirements and related test cases.

4.3.2 Technical baseline

There are no NWDAF-specific additions to clause 4.3.2 of TS 33.117 [3].

4.3.3 Operating systems

There are no NWDAF-specific additions to clause 4.3.3 of TS 33.117 [3].

4.3.4 Web servers

There are no NWDAF-specific additions to clause 4.3.4 of TS 33.117 [3].

4.3.5 Network devices

There are no NWDAF-specific additions to clause 4.3.5 of TS 33.117 [3].

4.3.6 Network functions in service-based architecture

There are no NWDAF-specific additions to clause 4.3.6 in TS 33.117 [3].

4.4 NWDAF-specific adaptations of basic vulnerability testing requirements and related test cases

4.4.1 Introduction

There are no NWDAF specific addtions to clause 4.4.1 of TS 33.117 [3].

4.4.2 Port scanning

There are no NWDAF specific addtions to clause 4.4.2 of TS 33.117 [3].

4.4.3 Vulnerability scanning

There are no NWDAF specific addtions to clause 4.4.3 of TS 33.117 [3].

4.4.4 Robustness and fuzz testing

The test cases under clause 4.4.4 of TS 33.117 [3] are applicable to NWDAF.

The interface defined for the NWDAF are in 4.2.3 of TS 23.501 [5].

According to clause 4.4.4 of TS 33.117 [3], the transport protocols available on the interfaces providing IP-based protocols need to be robustness tested. Following TCP/IP layer model and considering all the protocols over transport layer, for NWDAF, the following interface and protocols are in the scope of the testing:

- For Nnwdaf: the TCP, HTTP2 protocols and the format of JSON.

NOTE: There could be other interfaces and/or protocols requiring testing under clause 4.4.4 of TS 33.117 [3]

Annex A (informative): Change history

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2021-06	SA#92e	SP-210427				Presented for information and approval	1.0.0
2021-06	SA#92e					EditHelp review and upgrade to change control version	17.0.0
2021-09	SA#93e	SP-210898	0002	1	F	Clarification on Data Masking on Integration Analysis	17.1.0
2022-06	SA#96	SP-220547	0003	-	F	Delete Use Case on Finding the right NF instance are serving the UE	17.2.0
2023-06	SA#96	SP-230677	0004	1	B	Robustness interfaces and protocols defined for NWDAF	18.0.0
2025-10	-	-	-	-	-	Update to Rel-19 version (MCC)	19.0.0
2026-03	SA#111	SP-260161	0006	-	F	Update requirement and test case of clause 4.2.1.2.6 and add protocols to the robustness and fuzz testing	20.0.0
2026-06	SA#112	SP-260498	0007	-	B	Fine-grained authorization for analytics services based on Analytics ID	20.1.0
2026-06	SA#112	SP-260498	0008	-	B	ML model asset access control in cross-NWDAF model provisioning	20.1.0